MA388 Sabermetrics: Lesson 19

Comparing Peak Trajectories - Part II

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Lesson Objectives

1. Map a quadratic model function over multiple players to compare career trajectories.
2. Identify which historical home run hitter had the highest estimated peak HR rate.
3. Recognize unusual trajectory shapes and hypothesize explanations (e.g., injuries, era effects).

Comparing Trajectories of Home Run Hitters

1. Find the ten players in baseball history who have hit the most career home runs.

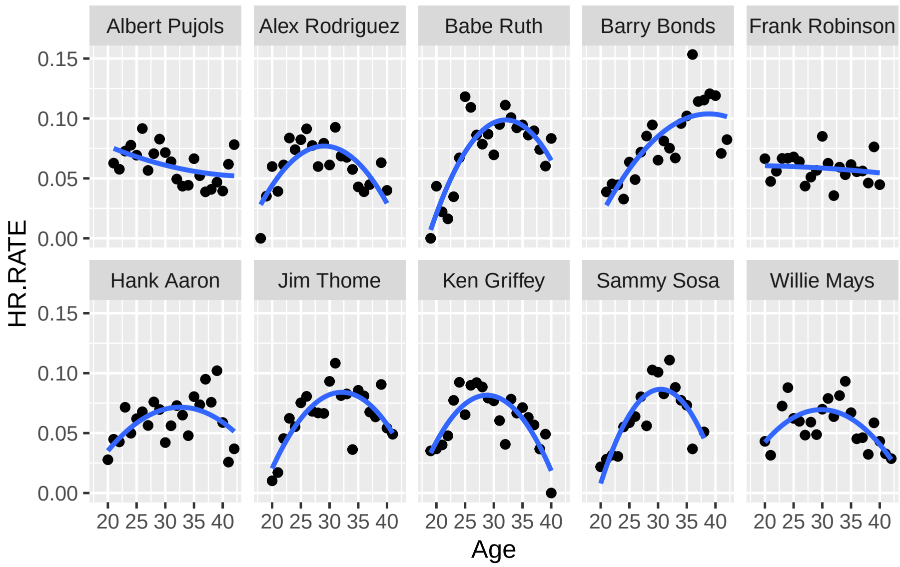
```
library(tidyverse)
library(Lahman)
library(knitr)
```

```
Top10 <- Batting |>
  summarize(totalHR = sum(HR), .by ='playerID') |>
  slice_max(totalHR,n=10) |>
  inner_join(People, by = "playerID") |>
  mutate(label = str_c(nameFirst,nameLast,sep=' ')) |>
  relocate(label,totalHR)

Top10 |>
  select(label, totalHR, debut) |>
  kable()
```

label	totalHR	debut
Barry Bonds	762	1986-05-30
Hank Aaron	755	1954-04-13
Babe Ruth	714	1914-07-11
Albert Pujols	703	2001-04-02
Alex Rodriguez	696	1994-07-08
Willie Mays	660	1951-05-25
Ken Griffey	630	1989-04-03
Jim Thome	612	1991-09-04
Sammy Sosa	609	1989-06-16
Frank Robinson	586	1956-04-17

2. Fit quadratic functions to the home run rates of the ten players, where HR.RATE = HR / AB. Display the fitted trajectories in a single figure.

```r
top10_stats <- Batting |>
  filter(playerID %in% Top10$playerID) |>
  summarize(HR.RATE = sum(HR)/sum(AB),.by=c('playerID','yearID')) |>
  inner_join(People, by = "playerID") |>
  mutate(birthyear = ifelse(birthMonth >= 7,
                            birthYear + 1, birthYear),
         label = str_c(nameFirst,nameLast,sep=' '),
         Age = yearID - birthyear)

top10_stats |>
  ggplot(aes(x = Age, y = HR.RATE)) +
  geom_point() +
  geom_smooth(method = "lm", formula = y~poly(x,2), se = FALSE) +
  facet_wrap(~label, ncol = 5)
```



3. On the basis of your work, which player had the highest estimated home run rate at his peak? Which player among the ten had the smallest peak home run rate?

```
top10_model <- function(this_playerID){

  this_player_stats <- top10_stats |>
    filter(playerID == this_playerID)

  fit <- lm(HR.RATE ~ I(Age - 30) + I((Age - 30)^2), data = this_player_stats)

  A <- coef(fit)[1]
  B <- coef(fit)[2]
  C <- coef(fit)[3]

  # Return all the stuff you just obtained.
  tibble(label = head(this_player_stats$label,1), playerID = this_playerID, A, B, C,
         age.max = 30 - B / C / 2,
         max = A - B^2 / C / 4)
}

# Map the model function to the list of playerIDs you obtained in 1.
HR_results <- map_df(Top10$playerID,top10_model)
HR_results |>
  kable()
```

label	playerID	A	B	C	age.max	max
Barry Bonds	bondsba01	0.0851324	0.0042375	-0.0002401	38.82412	0.1038285
Hank Aaron	aaronha01	0.0700492	0.0011897	-0.0002294	32.59352	0.0715920
Babe Ruth	ruthba01	0.0965013	0.0022230	-0.0005377	32.06727	0.0987990
Albert Pujols	pujolal01	0.0610317	-0.0012091	0.0000387	45.60271	0.0515987
Alex Rodriguez	rodrial01	0.0766162	-0.0007482	-0.0003996	29.06393	0.0769664
Willie Mays	mayswi01	0.0695671	-0.0000923	-0.0002813	29.83585	0.0695747
Ken Griffey	griffke02	0.0808547	-0.0012131	-0.0005022	28.79225	0.0815873
Jim Thome	thomeji01	0.0820880	0.0018386	-0.0004322	32.12711	0.0840435
Sammy Sosa	sosasa01	0.0864312	0.0006562	-0.0007198	30.45580	0.0865808
Frank Robinson	robinfr02	0.0586096	-0.0002988	-0.0000104	15.61690	0.0607582

```r
# Highest estimated max HR rate:
HR_results |>
  slice_max(max)
```

```
# A tibble: 1 x 7
  label       playerID       A       B         C age.max   max
  <chr>       <chr>      <dbl>   <dbl>     <dbl>   <dbl> <dbl>
1 Barry Bonds bondsba01 0.0851 0.00424 -0.000240    38.8 0.104
```

```r
# Lowest estimated max HR rate:
HR_results |>
  slice_min(max)
```

```
# A tibble: 1 x 7
  label         playerID       A        B         C age.max    max
  <chr>         <chr>      <dbl>    <dbl>     <dbl>   <dbl>  <dbl>
1 Albert Pujols pujolal01 0.0610 -0.00121 0.0000387    45.6 0.0516
```

Your answer here.

4. Do any of the players have unusual career trajectory shapes? Is there any possible explanation for these unusual shapes?

Your answer here.